

Task 1: Genomic Variants

COURSE NAME

CSDS 413 Introduction to Data Analysis

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Context

In this exercise, our aim is to quantify the association between genomic variants in the plant *Arabidopsis Thaliana* while assessing the statistical significance of their association. In this context, the variables are genomic variants and the samples are individual plants. We will use the two datasets that are provided with the assignment:

- The file "p1a.csv" contains a binary matrix of size 199 x 2 which indicates the presence of two genomic variants for 199 individuals. A value of 1 indicates that the genomic variant is present for the corresponding individual, and 0 indicates it is not.
- The file "p1b.csv" contains a binary matrix of size 199 x 15 which has 199 samples (individuals) and 15 variables (genomic variants). Each column represents a variable, and each entry (1 or 0) in the column indicates the presence of the corresponding variable in each of the 199 samples.

1 Part (a)

For the two variables provided in "p1a.csv", assess the association between them by computing each of the following statistics:

- Mutual Information
- Jaccard Index
- Pearson's chi-squared χ^2

For each of these statistics, assess the statistical significance by computing a p-value for the null hypothesis of "there is no association" (see *Note (1)* below). Select a significance level α that we can use to reject the null hypothesis and accept the alternate hypothesis "there is a non-zero association." if a p-value is less than α (see *Note (2)* below for the selection of α). For each of the three specified statistics, explain, in complete sentences, your findings: Is there a statistically significant association between the given variables? How strong does the association seem? Do the results of different statistics agree? If not, what does this mean (i.e., how do you explain this discrepancy)? Make sure to specify the values of the test statistics and the p-values as well as the selected α level and the number of permutations N you use for the permutation tests.

Note (1): The p-value in this context indicates the probability of observing a "more significant" result (for a statistic of interest like mutual information) than the observed result when there is no association between the genomic variants (which is our null hypothesis).

Note (2): You are welcome to select the commonly used 0.05 as your α threshold. However, you are encouraged to try different α values and observe how the conclusions of a researcher might change dramatically if they are not careful with the interpretations of the hypothesis test results.

Hint (1): For mutual information and Jaccard index, you need to do a Monte Carlo simulation (e.g., a permutation test) to assess statistical significance. For Pearson's χ^2 , you can use the parametric distribution (i.e., [Pearson's chi-squared test](#)) instead of doing a Monte-Carlo simulation to generate the null distribution.

Hint (2): The idea of the permutation test is to randomly permute the entries in each column (variable) separately, in order to break any possible associations between the variables without modifying their distributions. This way, we can generate a null distribution for the test statistic (like mutual information) that represents our null hypothesis. Performing the permutation test is simple: First, generate N random permutations of the data and for each of these permuted data, compute the test statistic of interest. Then, count the number of times permuted data has more significant values (e.g., higher values for mutual information) than the actual data, let's denote this count c . Then, the p-value is simply equal to $\frac{c+1}{N+1}$. Notice that, a p-value obtained from a permutation test cannot be lower than $\frac{1}{N+1}$.

Hint (3): An alternative to a permutation test (or Monte-Carlo simulation in general) is to obtain a null distribution analytically by fitting a known parametric distribution. For example, it is known that Pearson's χ^2 follows a well-studied distribution called [chi-squared distribution](#) when the null hypothesis of "no association" is correct. Thus, it is possible to obtain a p-value analytically from the chi-squared distribution (indeed, this is what the [Pearson's chi-squared test](#) does).

Hint (4): To visualize the results of a permutation test, you can draw a sorted scatter plot while marking the observed value (as shown in 'example-figure1.jpg').

To perform the permutation test framework on the p1a dataset for mutual information and Jaccard index, I set $N = 10,000$. For Pearson's χ^2 , I decided to follow the suggestion in the instruction and analytically fit the parametric chi-squared distribution with 1 degree of freedom.

For all three tests, I used $\alpha = 0.05$ to reject the null hypothesis, below is the implementation of the three statistics and the relevant tests:

```
def mutual_information(x: np.ndarray, y: np.ndarray) -> float:
    """
    Calculates mutual information between two binary variables.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
```

```

:type y: np.ndarray
:returns: Mutual information value
:rtype: float
"""
contingency = contingency_table(x, y)
n = len(x)
p_joint = contingency / n

p_x = p_joint.sum(axis=1)
p_y = p_joint.sum(axis=0)

mi = 0
for i in range(2):
    for j in range(2):
        if p_joint[i, j] > 0:
            mi += p_joint[i, j] * np.log2(p_joint[i, j] / (p_x[i] * p_y[j]))

return mi

def jaccard_index(x: np.ndarray, y: np.ndarray) -> float:
    """
    Calculates Jaccard Index between two binary variables.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :returns: Jaccard Index value
    :rtype: float
    """
    intersection = np.sum((x == 1) & (y == 1))
    union = np.sum((x == 1) | (y == 1))

    if union == 0: return 0

    return intersection / union

def chi_squared_statistic(x: np.ndarray, y: np.ndarray) -> float:
    """
    Calculates Pearson's chi-squared statistic.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :returns: Chi-squared statistic
    :rtype: float
    """
    contingency = contingency_table(x, y)
    chi2_stat, _, _, _ = chi2_contingency(contingency)
    return chi2_stat

def contingency_table(x: np.ndarray, y: np.ndarray) -> np.ndarray:
    """
    Creates 2x2 contingency table for two binary variables.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :returns: 2x2 contingency table

```

```

:rtype: np.ndarray
"""
table = np.zeros((2, 2))
table[0, 0] = np.sum((x == 0) & (y == 0))
table[0, 1] = np.sum((x == 0) & (y == 1))
table[1, 0] = np.sum((x == 1) & (y == 0))
table[1, 1] = np.sum((x == 1) & (y == 1))
return table

def permutation_test(x: np.ndarray, y: np.ndarray, statistic_func,
                    n_permutations: int) -> tuple:
    """
    Performs permutation test to calculate p-value for given statistic.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :param statistic_func: Function to calculate test statistic
    :type statistic_func: function
    :param n_permutations: Number of permutations
    :type n_permutations: int
    :returns: Tuple of observed statistic, p-value, null distribution
    :rtype: tuple
    """
    observed_stat = statistic_func(x, y)

    null_dist = np.zeros(n_permutations)

    for i in range(n_permutations):
        y_permuted = np.random.permutation(y)
        null_dist[i] = statistic_func(x, y_permuted)

    c = np.sum(null_dist >= observed_stat)
    p_value = (c + 1) / (n_permutations + 1)

    return observed_stat, p_value, null_dist

def chi_squared_parametric(x: np.ndarray, y: np.ndarray) -> tuple:
    """
    Performs Pearson's chi-squared test using parametric distribution.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :returns: Tuple of chi-squared statistic, p-value
    :rtype: tuple
    """
    contingency = contingency_table(x, y)
    chi2_stat, p_value, _, _ = chi2_contingency(contingency)

    return chi2_stat, p_value

```

Figure 1: Core functions for computing statistics and performing permutation tests.

The computed test statistics and corresponding p-values:

Statistic	Value	P-value
Mutual Information	0.047	0.0027
Jaccard Index	0.000	1.0000
Pearson's χ^2	6.456	0.0111

Figure 2: Test statistics and p-values for the two genomic variants in p1a.csv.

I didn't see a example-figure1.jpg anywhere for reference, but a sorted scatter plot sounds straightforward enough for MI and JC, below are the plots for each, with each of the observed statistics marked with a dashed red line.

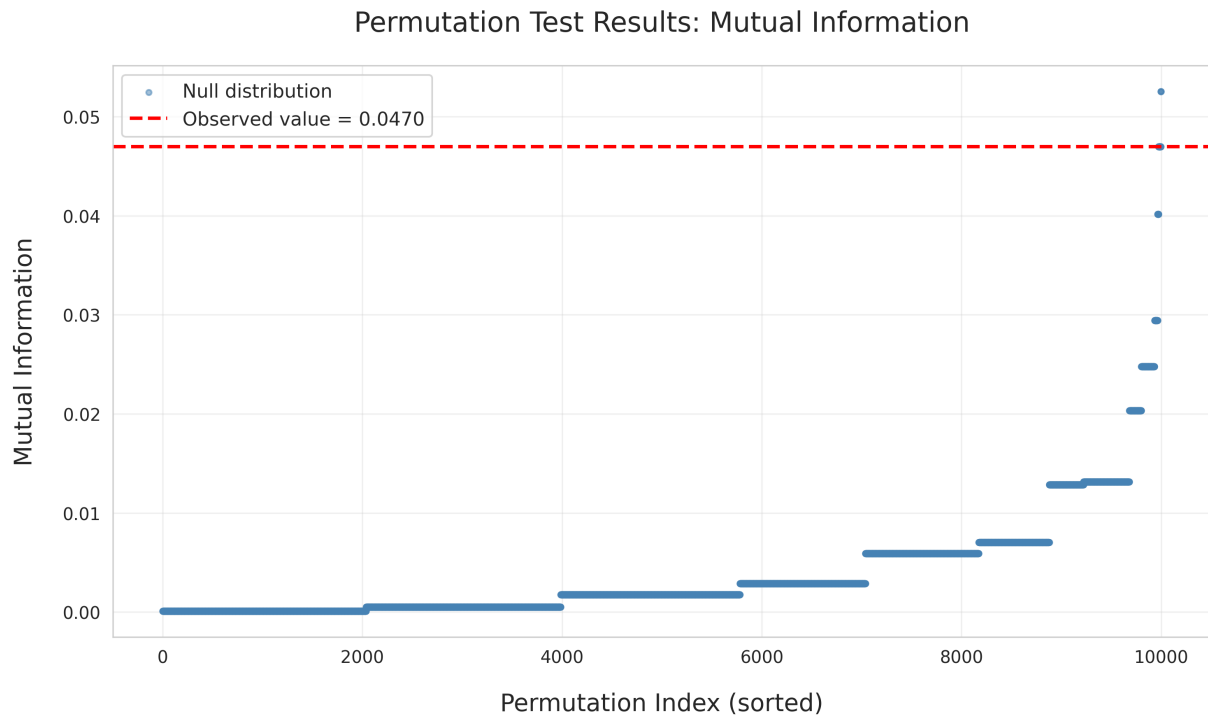


Figure 3: Permutation test results for Mutual Information.

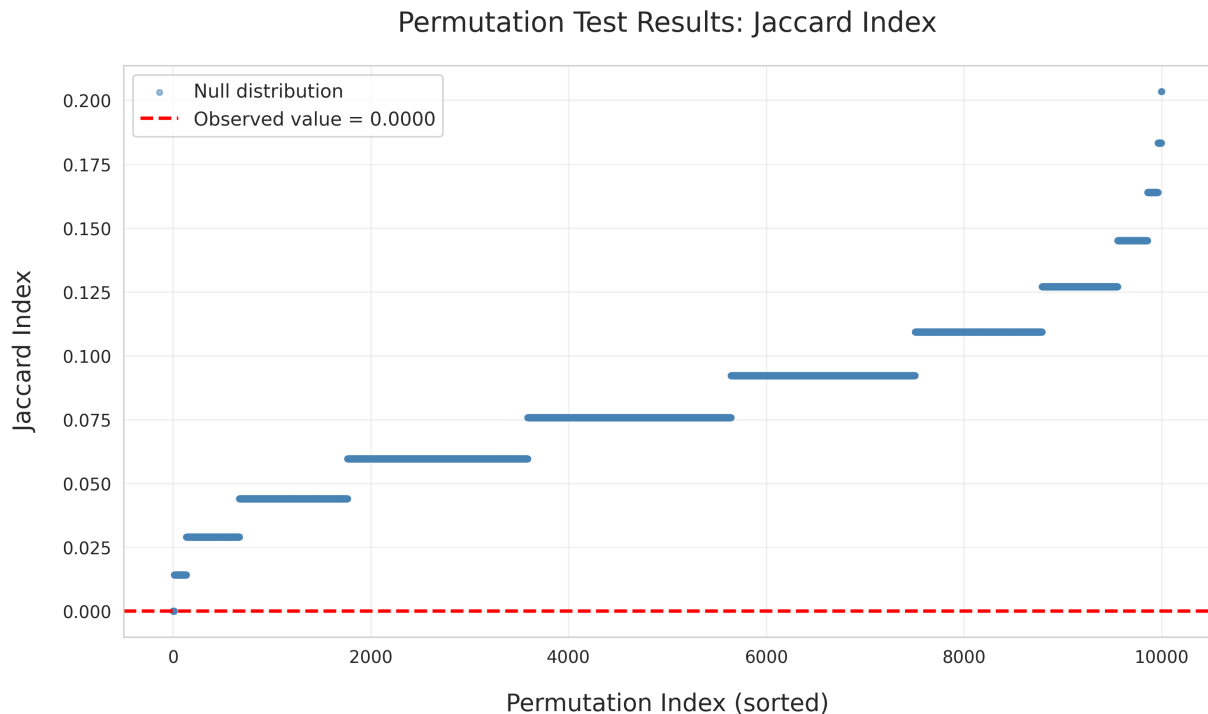


Figure 4: Permutation test results for Jaccard Index.

The MI of 0.047 is small, indicating that there is a small, but statistically detectable ($p = 0.0027$), amount of information shared between the variables; they are just a bit telling of each other. The chi-squared also shows a weak yet significant ($p = 0.0111$) association between the two variables with a value of 6.456, varying from what we ought to expect under independence. The Jaccard on the other hand, shows no association and upon inspection of the data following this result, I realized there is no occurrence of (1,1) in the data, yielding a value

of 0 and a p value of 1.

This means the statistics do not all agree, but there is good reason for this; it's a demonstration of the fact that these statistics capture different aspects of association between two variables, and in this specific case, we have a set of data that doesn't align well necessarily with how these statistics are defined. Jaccard is specifically helpful in the case where there is a relation between the (1,1) class and the union in the contingency table, but in this example, it is likely that the detection of any association is a result of the samples being mutually exclusive rather than the variables ever both being 1 at the same time. For this reason, MI and χ^2 are equipped to identify the occurrence of one variable given the result of another and the difference between the expected and observed values under independence, while the Jaccard is left blind to the connection by its construction.

2 Part (b)

For each of the 105 ($15 \times 14/2$) pairs of variables provided in "p1b.csv", repeat the steps in part (a) to compute the values of the test statistics as well as the p-values separately for: Mutual information, Jaccard Index and Pearson's chi-squared χ^2 statistics. Select an α level to reject the null hypothesis of no association. For each of the three test statistics, determine the pairs with significant association at α level while adjusting for false discovery rate (see Note (3) below). Draw scatter plots (or other appropriate visualizations) to compare the results of mutual information (MI), Jaccard Index (JI) and Pearson's chi-squared χ^2 . How much do the results of these three test statistics agree to one another? Which two of these three statistics are most similar (MI-JI, MI- χ^2 , or JI- χ^2)? If you were to use only one of these three statistics, which one would you prefer to use and why? If you were to consider only your preferred test statistic among the given three, would your conclusions (about the associations of the provided variable pairs) change a lot? Make sure to specify the number of significantly associated pairs and the number of overlap between the three test statistics. Also, specify the selected α level and the number of permutations N you use for the permutation tests.

Note (3): Here, differently from part (a), we are testing for multiple hypotheses (one for each variable pair). Therefore, if we were to reject a null hypothesis as suggested in part (a) (when p-value is less than α), we would have a considerably higher **false discovery rate (FDR)** than α (Think: If each test has a α chance of failure, what is the probability that there is at least one test that fails? For $\alpha = 0.05$ and $n = 105$ tests, that would be 99.54% chance of failure!). Thus, we need to use an appropriate procedure that take into account the number of hypotheses to make sure the FDR is indeed less than α . For this purpose, you can apply one of the following two approaches: (1) **Benjamini-Hochberg** procedure (which is, in many fields, the go-to way of adjusting for multiple hypotheses) to determine the statistically significant pairs, (2) A Monte Carlo simulation specifically designed to control the FDR based on the distributions generated by the permutations (e.g., to compute the p-value of the top-scoring pair, we can use the distribution of the test statistics of the top-scoring pairs across all permutations, which would be a "multiple-hypothesis-corrected" null distribution).

Note (4): Using the Benjamini-Hochberg procedure, essentially a stricter threshold is used to reject a null hypothesis (e.g., the threshold of $\frac{\alpha}{n}$ for rejecting the null hypothesis of the most significant pair). Therefore, for estimating the p-values using permutation test, you need to use more permutations compared to part (a) to be able find a significant association. Thus, make sure that your number of permutations N is high enough to be able to find at least one significant pair.

For the purpose of testing for multiple hypotheses and controlling for FDR, I applied the Benjamini-Hochberg (BH) procedure at $\alpha = 0.05$: to identify the most significant associations after FDR correction, we need enough permutations to reliably estimate p-values at about $0.05/105 \approx 0.0005$; towards this end, I set $N = 50,000$. Figure 5 gives the number of significant pairs detected by each statistic and Figure 6 gives the overlap between them:

Statistic	Significant Pairs
Mutual Information	91
Jaccard Index	56
Pearson's χ^2	90

Figure 5: Number of significant pairs detected by each statistic after Benjamini-Hochberg FDR correction at $\alpha = 0.05$.

Comparison	Number of Overlapping Pairs
MI $\cap$ JI	55
MI $\cap$ χ^2	89
JI $\cap$ χ^2	56
MI $\cap$ JI $\cap$ χ^2	55

Figure 6: Overlap of significant pairs between different statistics.

Below are the scatter plots side-by-side comparing each of the statistics:

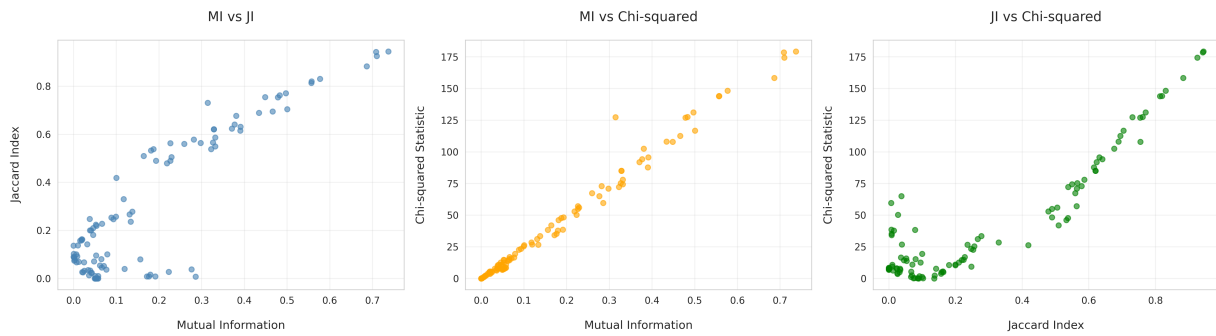


Figure 7: Pairwise comparison of test statistics across 105 genomic variant pairs.

To characterize each pair from the visual: MI and chi-squared are very similar, showing a very high level of agreement. Jaccard relative to the other two show similar and moderate levels of agreement, but both show some level of difference at lower extremes.

Unsurprisingly from the discussion in the previous part, MI and chi-squared show a high degree of agreement while each agrees somewhat with Jaccard. Again, because both of these statistics use all four cells of the contingency table to represent deviation from independence, and Jaccard is constructed more as a similarity measure that is blind to the mutual exclusivity pattern in the data, their ability to identify associations in this data are both better than Jaccard and most similar to each other.

I would say that using either MI or Pearson's chi-squared is going to generally arrive at the same result; the only case where just using one statistic would get you very different conclusion would be Jaccard. However, the computational efficiency of using chi-squared and not having to do many permutation tests to get essentially the same result would make it my preferred statistic. I think the difference in significant pairs by one may be more telling with additional exercises, but its not compelling enough given this test to believe that MI is noticeably superior.

Task 2: Continuous Variables

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Context

In this exercise, our aim is to quantify the associations between continuous variables and assess the statistical significance of these associations. For this purpose, we will use the three datasets that are provided with the assignment:

- The file 'p2a.csv' contains a matrix of size 2400 x 2 which has 2400 samples and 2 variables.
- The file 'p2b.csv' contains a matrix of size 110 x 2 which has 110 samples and 2 variables.
- The file 'p2c.csv' contains a matrix of size 2100 x 2 which has 2100 samples and 2 variables.

1 Part (a)

For the two variables provided in "p2a.csv", assess the association between them by computing Pearson correlation r_a and computing a p-value p_a for the null hypothesis of no association. Select a significance level α and reject the null-hypothesis if the p-value is less than α . Explain, in complete sentences, your findings: Is there a statistically significant association (at α level) between the provided variables? What is the magnitude and the direction of the association?

Below is the function and the tabulation for r_a and p_a :

```
def compute_pearson_correlation(x: np.ndarray, y: np.ndarray) -> tuple:
    """
    Computes Pearson correlation coefficient and p-value.

    :param x: First variable
    :type x: np.ndarray
    :param y: Second variable
    :type y: np.ndarray
    :returns: Tuple of (correlation coefficient, p-value)
    :rtype: tuple
    """
    r, p_value = pearsonr(x, y)
    return r, p_value
```

Figure 1: Function for computing Pearson correlation and p-value.

Pearson r	P-value
0.3809	1.04×10^{-83}

Figure 2: Pearson correlation results for p2a.

The Pearson correlation given by the function was 0.380875. Given a selected significance level $\alpha = 0.05$, there is a statistically significant, positive association between the two variables. The p value is incredibly low, magnitudes lower than the selected α , but 0.05 is convention and follows most of the class exercises we have done. By no means is this result challenged by that threshold, so we can confidently reject the null hypothesis.

2 Part (b)

Repeat part (a) for the variable pair provided in "p2b.csv" and compute Pearson correlation r_b and p-value p_b . Compare the Pearson correlations r_a and r_b as well as the p-values p_a and p_b . Explain your findings: Which variable pair (in part a or b) has a stronger association according to the comparison of the correlations? Which variable pair has a stronger association according to the comparison of the p-values? Do the comparisons according to correlation coefficients and p-values agree on which variable pair indicate the same stronger association? If not, why is there such a discrepancy? Next, draw scatter plots (variable 1 vs. variable 2) to visualize the data for both part (a) and part (b). Which variable pair (in part a or b) has a stronger association do you think according to the scatter plots? Does your conclusion agree with the comparisons of the p-values and correlation coefficients? If not, explain why.

Below is the tabulation of the Pearson correlations and p values as well as scatter plots for each dataset:

Dataset	Pearson r	P-value
p2a	0.3809	1.04×10^{-83}
p2b	0.9312	3.74×10^{-49}

Figure 3: Comparison of Pearson correlation results between p2a and p2b.

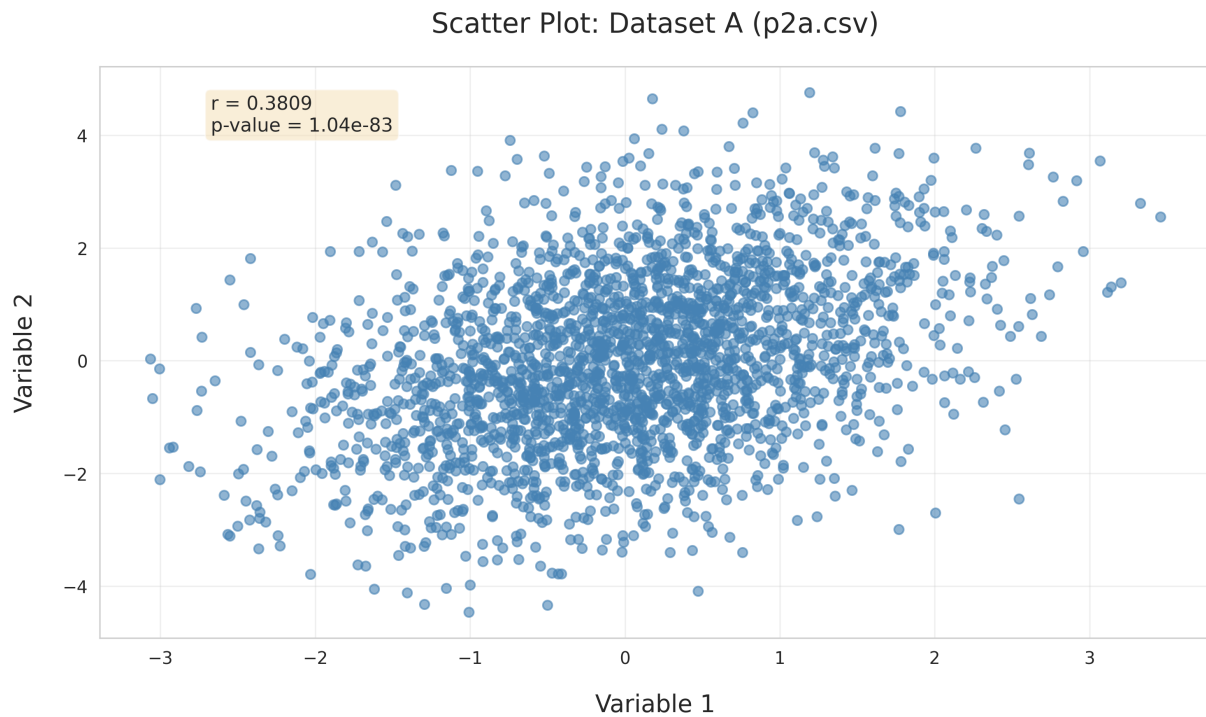


Figure 4: Scatter plot for p2a.csv.

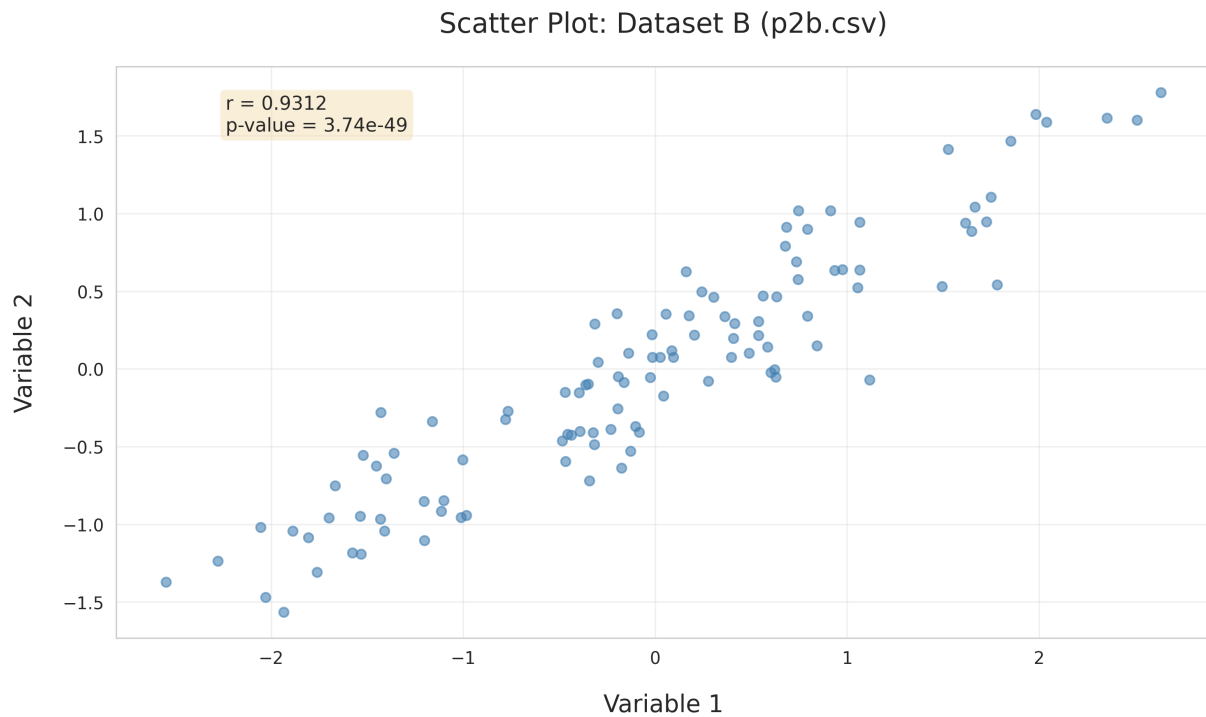


Figure 5: Scatter plot for p2b.csv.

The Pearson correlation for p2b came to 0.9312, and given the same significance level $\alpha = 0.05$, there is a statistically significant, positive association between the two variables. The p value $p_b = 3.74 \times 10^{-49}$ is again much lower than the threshold, so we can confidently reject the null hypothesis.

According to the raw correlations $r_a = 0.3809$ and $r_b = 0.9312$, the part b variable pair shows a stronger correlation. However, the p-values $p_a = 1.04 \times 10^{-83}$ and $p_b = 3.74 \times 10^{-49}$ suggest that the part a variable pair is much smaller and therefore shows a stronger association, so these criteria don't agree. The reason for this is the difference in statistical power that lies in the sample size. p2a contains 2400 samples, while p2b only has 110: so in the case of p2a, even a moderate correlation becomes much more statistically significant because there is much more evidence that the pattern is not occurring by chance, whereas p2b has much less data to rule out randomness at play contributing to the association, and so even such an extreme correlation cannot produce a lower p-value.

Regarding the scatter plots, I would say off visual inspection that p2b shows a much clearer relationship, albeit the graph is much more sparse. The data stick very closely to the trend line as you would expect for such a high correlation, whereas p2a shows some agreement generally in the positive direction but as a whole kind of bulges out and looks befitting of a sample that would yield a moderate correlation value. So, my visual assessment agrees with the correlation assessment and not the p-value assessment. The p-value in p2a is simply a product of a larger sample and ought not to be misconstrued as an indication of a stronger relationship. Not to mention that the strength of p-values does not lie in making claims about qualities of the data itself, but more so in the uncertainty we possess in such claims. For that reason, I think my visual assessment of the scatter plots makes sense.

3 Part (c)

Repeat part (b). But, this time compare the associations in the datasets "p2a.csv" and "p2c.csv". Make sure to answer the questions in part (b), this time for comparing the variable pairs in parts (a) and (c).

Below is the tabulation of the Pearson correlations and p values as well as scatter plots for each dataset:

Dataset	Pearson r	P-value
p2a	0.3809	1.04×10^{-83}
p2c	0.0412	0.0592

Figure 3: Comparison of Pearson correlation results between p2a and p2c.

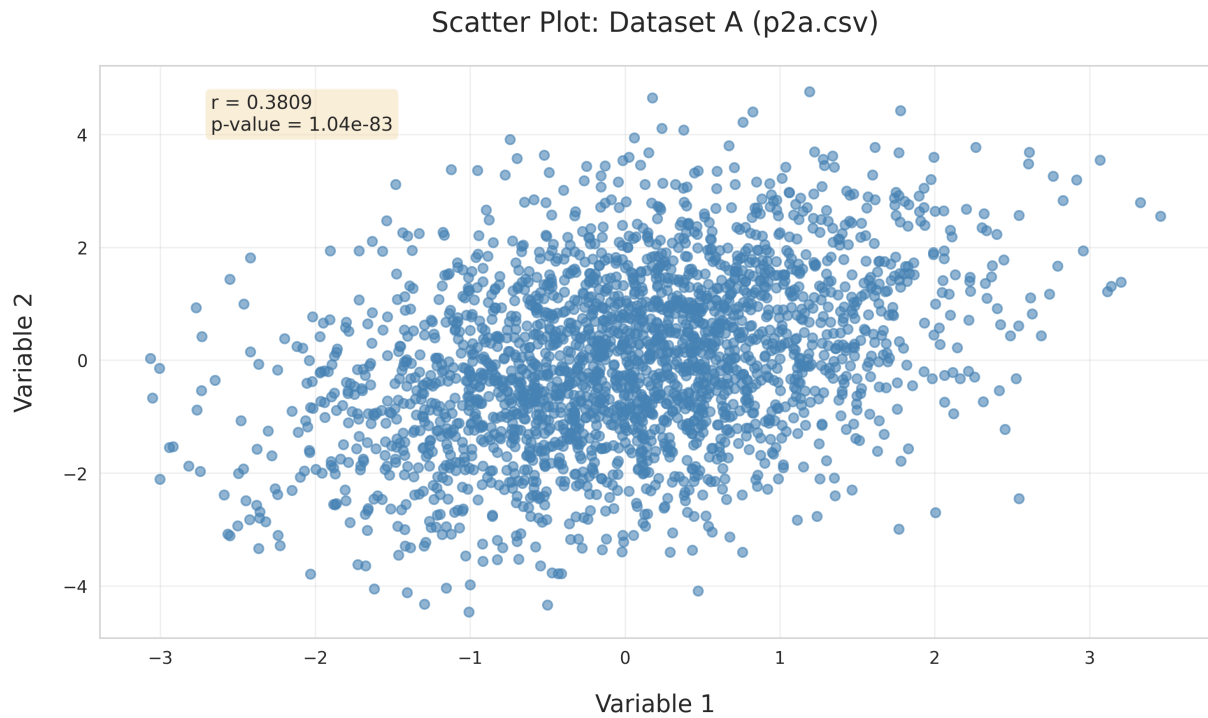


Figure 4: Scatter plot for p2a.csv.

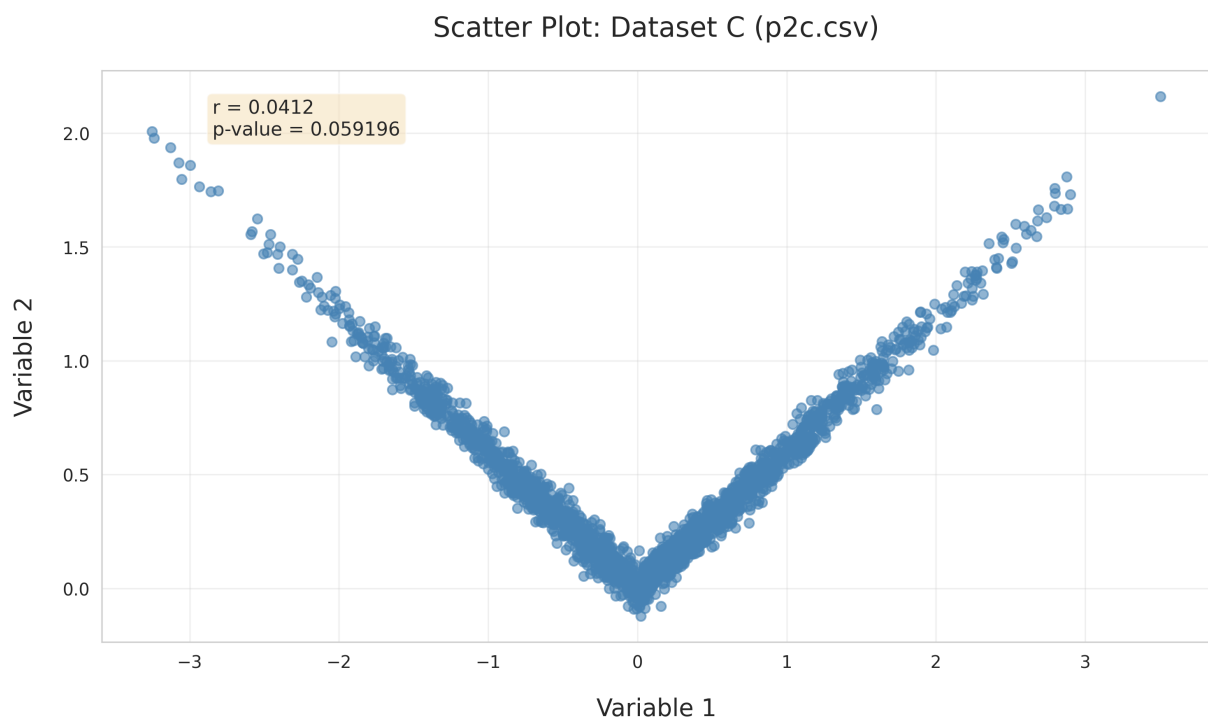


Figure 5: Scatter plot for p2c.csv.

The Pearson correlation for p2c came to 0.0412, and given the same significance level $\alpha = 0.05$, there is not a statistically significant association between the variables. Given the same significance level $\alpha = 0.05$, the p value $p_c = 0.0592$ is higher than the threshold, so we fail to reject the null hypothesis.

According to the raw correlations $r_a = 0.3809$ and $r_c = 0.0412$, the part a variable pair shows a stronger correlation. Similarly, the p-values $p_a = 1.04 \times 10^{-83}$ and $p_c = 0.0592$ suggest that the part a variable pair is much smaller and therefore shows a stronger association, so the criteria agree and there is no discrepancy.

As far as the scatter plots, again the p2a dataset shows a generally positive agreement that vaguely sticks to the trend line; a result you would expect from a moderate, positive Pearson correlation value. The p2c dataset however shows a very odd behavior: it seems as though as variable 1 increases and is negative, there is a very strong negative association conveyed by variable 2, and similarly for positive variable 1 values, variable 2 shows a very strong positive association. I would think that if p2c just possessed only negative variable 1 samples or only positive variable 1 samples, these results may be very different.

If we are to just go off of visual assessment, Pearson correlation aside and just considering the relationship between the variables, I would claim that the p2c variable pair shows much stronger association than p2a. This disagrees with both suggestions made by the Pearson correlations and the p-values. This exercise appears to be a demonstration of how Pearson correlation strictly quantifies linear associations and what scenarios it may fail in.

Interestingly enough, the occurrence of positive variable 1 samples and negative variable 1 samples is likely so even in p2c, the two relationships on each side of the y-axis almost perfectly cancel each other out, giving a correlation very close to zero, despite it being immediately apparent from the plot that a strong, non-linear association is present between the variables. That being said, the reason my visual assessment disagrees with the suggestions made by the Pearson correlations and p-values is that I can see from the scatter plot this non-linear relationship that Pearson correlation is blind to, and without the necessary nuance to identify it, the suggestions made by the Pearson correlations and the associated p-values are not compelling in this context.